

$$\begin{array}{ccc}
 (S, \mu) & \xrightarrow{g} & (S, \mu) \\
 \downarrow \phi & & \uparrow \phi^{-1} \\
 (X, \mu_0) & \xrightarrow{f = \phi \circ g \circ \phi^{-1}} & (X, \mu_0)
 \end{array}$$